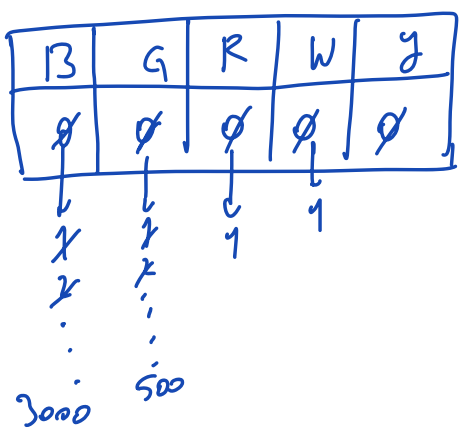
Country Sorts

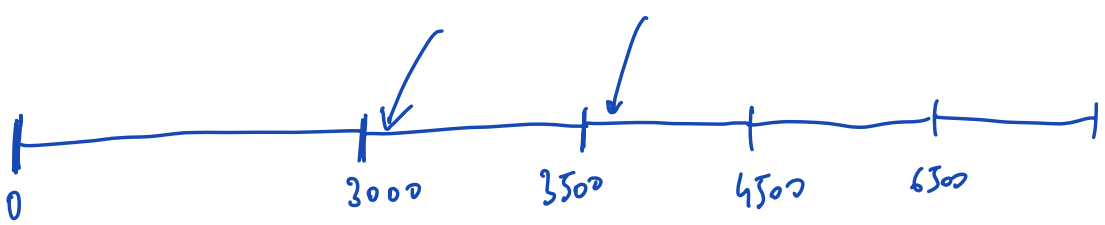
Car entries: Colors: White 2000
Blue 3000
Red 1000
Green 500
Yellow. 3500

10,000 car entries:

BBB . . . . . BGG . . . GRR . . . . RWW . . . . W YYY . . . Y



Color Table

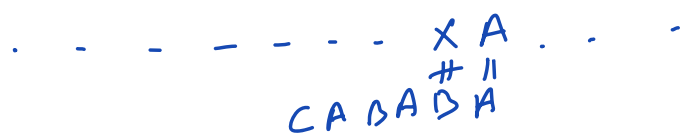


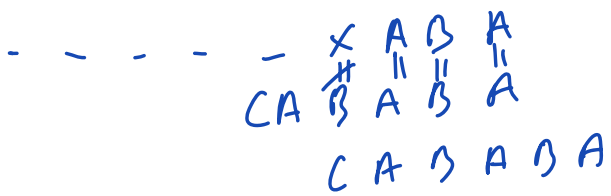
CABABA

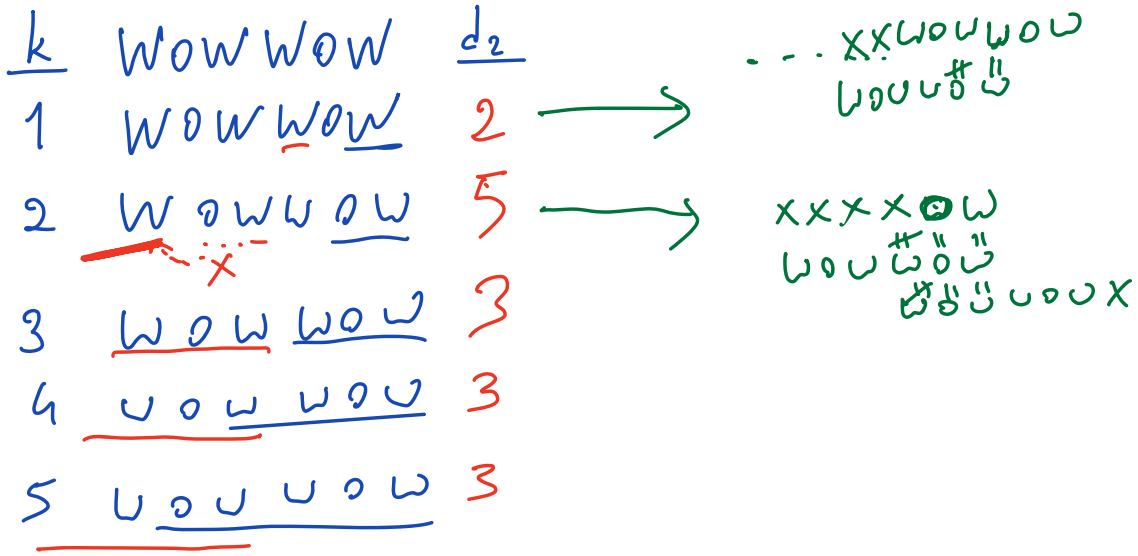
Bad Symbol table

B → 1
A → 2
C → 5
others → 6

k	C A B A B A	d_2
1	C A D A B A	4
2	C A D A B A	6
3	C A B A B A	2
4	C A D A D A	6
5	C A B A B A	6







What is the worst case # comparisons for Horspool?

text length: n
pattern length: m

n zeros

m checks.

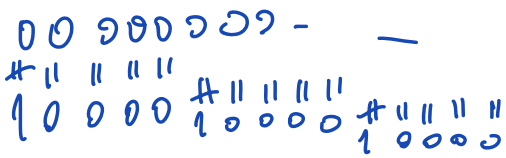
In the worst case # comparisons:

$m(n-m+1)$

Shift table:

1 → 4

0 → 1



$d_1 = \max\{1-4, 1\}$ $d_2 = 5$

$= 1$ $(k=4)$

if $n = 1000$

comparisons: 1000

Boyer moore:

Bad symbol table

$1 \to 4$

$0 \to 1$

Good suffix table:

		d_2
$k=1$	10000	3
$k=2$	10000	2
$k=3$	10000	1
$k=4$	10000	5

Consider a text with 1000 zeros.

How many char comparisons will be done by Boyer-moore, when we search 001010.

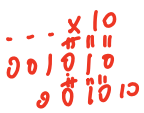
Bad symbol table:

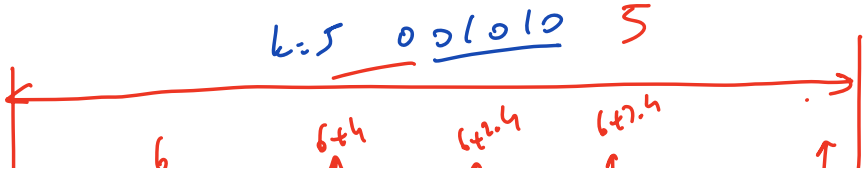
$1 \to 1$

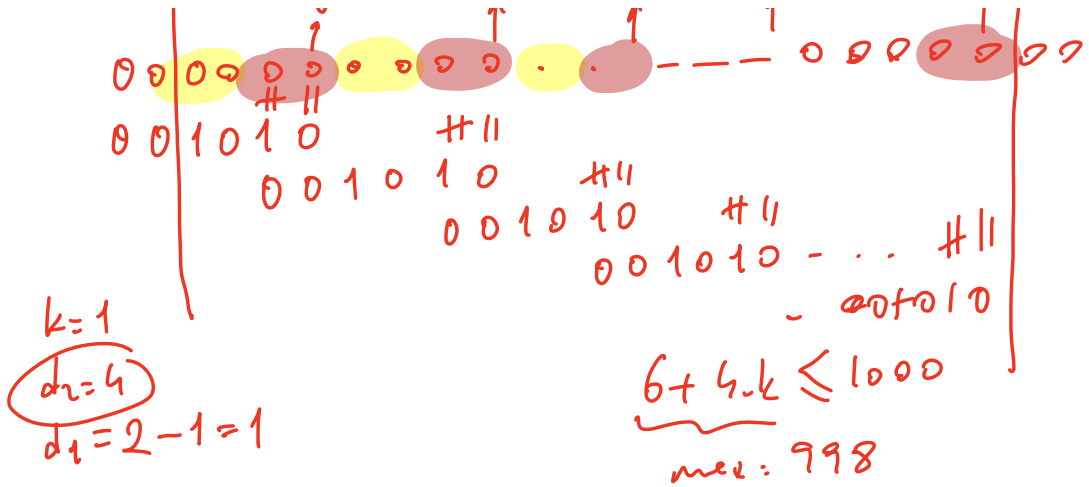
$0 \to 2$

Good suffix table

		d_2
$k=1$	001010	4
$k=2$	001010	5
$k=3$	001010	2
$k=4$	001010	5
$k=5$	001010	5







$\frac{996}{2}$ characters = 498 character expansions.